

# IMESH MALINDA

[jane-doe.com](#) | [LinkedIn](#) | [GitHub](#) | [Leetcode](#)

Location: Kahathuduwa, Colombo, Sri Lanka

Email: [imeshmalinda8626@gmail.com](mailto:imeshmalinda8626@gmail.com) | Mobile: 0756799270

## SUMMARY

Electrical and Information Engineering graduate with practical training experience in power generation, transmission, distribution, and building electrical services. Gained technical exposure through industrial training at the Ceylon Electricity Board (CEB) and Tudawe Engineering Services, including power plants, grid substations, transformer testing, electrical distribution systems, fire detection systems, and HVAC systems. Strong interest in power systems and electrical engineering, with good technical communication, teamwork, safety awareness, and problem-solving skills.

## TECHNICAL SKILLS

|             |                                                                                                                                                                                                                  |
|-------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Technical   | : Testing and Commissioning, Cable Sizing, Earthing Design, Electrical Installation, Lightning Protection, Fault Analysis, Load Flow Analysis, Single-Line Diagrams (SLDs), IEC Standards, Blockchain Technology |
| Softwares   | : OpenDss, DIALux evo, ETAP, AutoCAD Electrical, MATLAB/Simulink, Homer Pro, Proteus, Power World Simulator, MS Office                                                                                           |
| Soft Skills | : Leadership, Teamwork, Problem Solving, Communication, Time Management, Analytical Thinking                                                                                                                     |

## EXPERIENCE

### Trainee Electrical Engineer

Ceylon Electricity Board (CEB), Sri Lanka

Aug 2025 – Oct 2025 | [Service Letter](#)

- Gained practical knowledge of hydropower and coal power generation, including turbines, generators, switchyards, power transformers, and overall power plant operations.
- Gained technical exposure to grid substations, including circuit breakers, isolators, CTs, CVTs, protection systems, busbar arrangements, and DC battery systems.
- Gained practical knowledge of transformer testing, including insulation resistance and transformer oil breakdown voltage (BDV) testing.
- Gained exposure to electrical distribution systems, including energy meter testing, distribution construction, planning, development, and maintenance.

### Trainee Electrical Engineer

Tudawe Engineering Services (Pvt) Ltd.

May 2024 – Aug 2024 | [Service Letter](#)

No. 505/2, Elvitigala Mawatha, Colombo 05, Sri Lanka

- Gained practical experience in the implementation of fire detection and alarm systems for a large-scale building project.
- Assisted with the installation, inspection, and implementation of fire detection system components and related electrical works.
- Gained practical exposure to HVAC systems and their electrical requirements.

### Assistant Electrician

Sanota Constructions

May 2021 – Dec 2021 | [Service Letter](#)

No. 45/2/17, Gardenia Grow, kendaketiya, Kiriwaththuduwa.

- Assisted with domestic electrical installations, including wiring, installation of electrical accessories, and related electrical works.

## EDUCATION

### University of Ruhuna

B.Sc. Eng. (Hons) in Electrical and Information Engineering

Galle, Sri Lanka

Mar 2022– Jul 2026

- Overall GPA :3.55/4.00

- Course Work :Power Systems, Power System Analysis, Electric Machines, Electric Installations, High Voltage Engineering, Control System Design, Renewable Energy, Power System Protection

## **Mahinda Rajapaksha College**

G.C.E. Advanced Level

Homagama, Sri Lanka

2020

- Physics :A
- Chemistry :A
- Combined Mathematics :A
- Z-Score :1.8288

G.C.E. Ordinary Level

2017

- Passed with 9As

## **PROJECTS**

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### **Blockchain-Based Local Energy Markets for Increasing Solar PV Penetration and Electric Vehicle Integration in Low-Voltage Distribution Networks**

*Final Year Project*

- Modelled and analyzed a low-voltage distribution network using OpenDSS with solar PV, EVs, energy storage, and different load profiles.
- Modelled a Battery Energy Storage System using HOMER PRO software.
- Developed a P2P energy trading approach using blockchain-based smart contracts for day-ahead and intraday local energy markets.
- Evaluated the impact of solar PV and EV integration on reverse power flow, voltage fluctuations, and PV hosting capacity.

### **Electrical Installation Design for Apartment Building**

*May 2026*

- Designed a BS 7671-compliant electrical installation for a three-storey residential apartment building, including power, lighting, protection, and backup systems.
- Performed load assessment, cable sizing, voltage-drop calculations, protective-device selection, SLD preparation, and lighting design.

### **Lightning Protection System Design**

*Jul 2026*

- Designed an IEC 62305-compliant lightning protection system for a four-storey educational building using risk assessment, LPL III, and the rolling sphere method.
- Prepared AutoCAD layouts for air termination, down conductors, and earthing, including sizing of 12 down conductors and conductor selection.

### **Hostel Management System**

*Sep 2024 – Sep 2025*

- Developed a web-based hostel management platform to digitalize and streamline administrative activities within university hostels.
- Built the frontend using React.js and utilized Firebase as the backend database.
- Implemented secure student registration and login, hostel defect reporting, and a virtual notice board for announcements.
- Developed an interactive administrator dashboard for monitoring hostel activities and managing information.

### **Sustainable Irrigation System**

*Mar 2025 – Jul 2025*

- Designed and developed an IoT-based solar-powered irrigation system integrating soil moisture sensing and automated irrigation for efficient water management.
- Implemented a sun tracking solar panel mechanism to maximize energy capture and support continuous system operation.
- Integrated real time monitoring and remote control through a web interface for smart and efficient irrigation management.

## Indoor Air Quality Monitoring System

Oct 2024 – Jan 2025

- Designed and developed an IoT-based indoor air quality monitoring system using Arduino, ESP8266, DHT11, and MQ-135 sensors.
- Implemented real-time monitoring of temperature, humidity and air pollutants
- Developed wireless data transmission and a web-based interface for remote monitoring and visualization of indoor environmental conditions.

## Highway Bus Seat Reservation System

Oct 2023 – Dec 2023

- Developed a C++ console-based bus seat reservation system for booking, viewing, and cancelling reservations.
- Applied OOP concepts including classes, objects, arrays, and functions.
- Implemented seat availability and double-booking prevention.

## SM Tracker – Social Media Usage Tracker

2022

- Developed a mobile application to track users' social media usage and screen time across different platforms.
- Designed the system to increase awareness of digital habits and encourage healthier and more balanced social media usage.

## CERTIFICATIONS

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- Certified Web Developer by the W3C
- Microsoft Certified: Azure Developer Associate
- AWS Certified Developer - Associate